Murad Hossain

Smart Manufacturing & Industry 4.0 | Research-focused academic profile

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Research direction

My industrial work connects sensor-based monitoring, manufacturing traceability, fixture improvement and engineering analysis to repeatable quality and production decisions.

Education and English qualification

BSc in Mechanical Engineering — Rajshahi University of Engineering & Technology (RUET), 2019. IELTS Academic overall 7.5 (Listening 8.5, Reading 7.5, Writing 6.5, Speaking 6.5).

Track-relevant engineering evidence

- Pirani gauge / CAN-HS vacuum process monitoring: Sensor and process-system implementation. Industrial monitoring implementation.
- Cooling-circuit evacuation process improvement to reduce choking-related issues: Process evaluation and implementation. Reported choking issue rate 0.13% to 0.02% across 2021–2025; multiple interventions were involved.
- Manufacturing barcode traceability cross-check: Workflow and process engineering. Traceability workflow; independent defect-reduction metric not supplied.
- Image tracking and finished-product records: Process integration. Industrial tracking implementation.
- Assembly-line balancing and SAM time study: Production engineering. Process analysis and operator/workstation allocation.

Research records

- FEA-Driven Topology Optimization for Lightweight Structural Design: A Computational Framework with Implicit Geometry Reconstruction — Accepted conference paper · ICMIME 2026 · ID 41.

Relevant credentials

Six Sigma Yellow Belt · SOLIDWORKS CAD Design Associate · 3DSwymer Associate · Python Coder

Technical skills

Quality analytics · Traceability · CAN-HS · Process development · SolidWorks · Time study

Employment context

Walton Hi-Tech Industries PLC (May 2021–present); New Model Boiler Company Limited (Nov 2019–Apr 2021). Full employment records and references available on request.